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A DBMS Project Report on

ALASTKA - Country-Based Tourism Information System

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1.INTRODUCTION

ALASTKA is a country-based tourism information system created to collect, organize, and manage tourism data for different countries and cities. It provides structured information about tourist destinations, attractions, and bookings, allowing users to explore places easily through one central platform.

Today, most travelers depend on online systems to plan trips, compare destinations, and find reliable travel information. However, tourism data is often spread across many websites and sources, which makes it hard to find complete and well-organized details in one place. ALASTKA solves this problem by offering a single, database-driven system that stores and retrieves tourism information in an organized and efficient way.

The main goal of this project is to show how Database Management System (DBMS) concepts work in real-life applications. It demonstrates important ideas such as entity-relationship modeling, relational database design, and normalization. The system supports key features like browsing destinations, categorizing tourist places, and managing bookings while maintaining data accuracy and consistency.

This project focuses on building a scalable and well-structured database that makes tourism data easier to manage and improves the overall user experience. ALASTKA also serves as an academic project that helps students understand how DBMS concepts are applied in practical scenarios.

The name **ALASTKA** is inspired by Alaska and Chukotka. Although these regions look far apart on flat world maps, they are actually very close geographically. This idea represents map projection illusions and reflects the project's purpose: connecting distant locations through a unified tourism information system.

2.PROBLEM STATEMENT

Tourism information is usually spread across many different websites and platforms. Because of this, travelers often find it hard to get accurate, complete, and well-organized details about destinations, tourist attractions, and accommodations. The absence of a single centralized system leads to inconsistent information, slow searching, and difficulty in planning trips efficiently.

In many cases, tourism data is managed manually or stored in unorganized digital formats. This can cause repeated data entries, outdated information, and errors in records. Maintaining and updating such data becomes time-consuming and unreliable. Therefore, there is a strong need for a structured, database-based system that can store, manage, and retrieve tourism information in an efficient and organized way across multiple countries and cities.

The objective of this project is to design and develop a Country-Based Tourism Information System that acts as a centralized platform for tourism data management. The system organizes destination details, categorizes tourist locations, and manages booking records while ensuring data accuracy, consistency, and easy access. By applying core Database Management System (DBMS) concepts, this project aims to create a scalable and well-structured solution that improves both data management and the overall user experience.

3. SYSTEM ARCHITECTURE AND MODULES

The ALASTKA Country-Based Tourism Information System follows a client–server architecture, where users interact with the system through a user interface while all tourism-related data is stored and managed in a centralized database. The system is designed to ensure efficient data storage, retrieval, and management using structured database operations.

The architecture consists of three main components: the user interface layer, application logic layer, and database layer. The user interface allows users and administrators to interact with the system. The application logic processes user requests and performs validations, while the database layer stores information related to countries, cities, tourist places, users, and bookings.

The system is divided into the following functional modules:

1. User Management Module

Handles user registration, login, and profile management. It ensures secure access to the system for both users and administrators.

2. Country Management Module

Allows administrators to add, update, and manage country information within the database.

3. City Management Module

Manages city details associated with each country and maintains their relationships.

4. Tourist Place Management Module

Stores and categorizes tourist attractions such as historical, natural, and adventure places.

5. Booking Module

Maintains booking records created by users and stores related information such as booking date and selected tourist place.

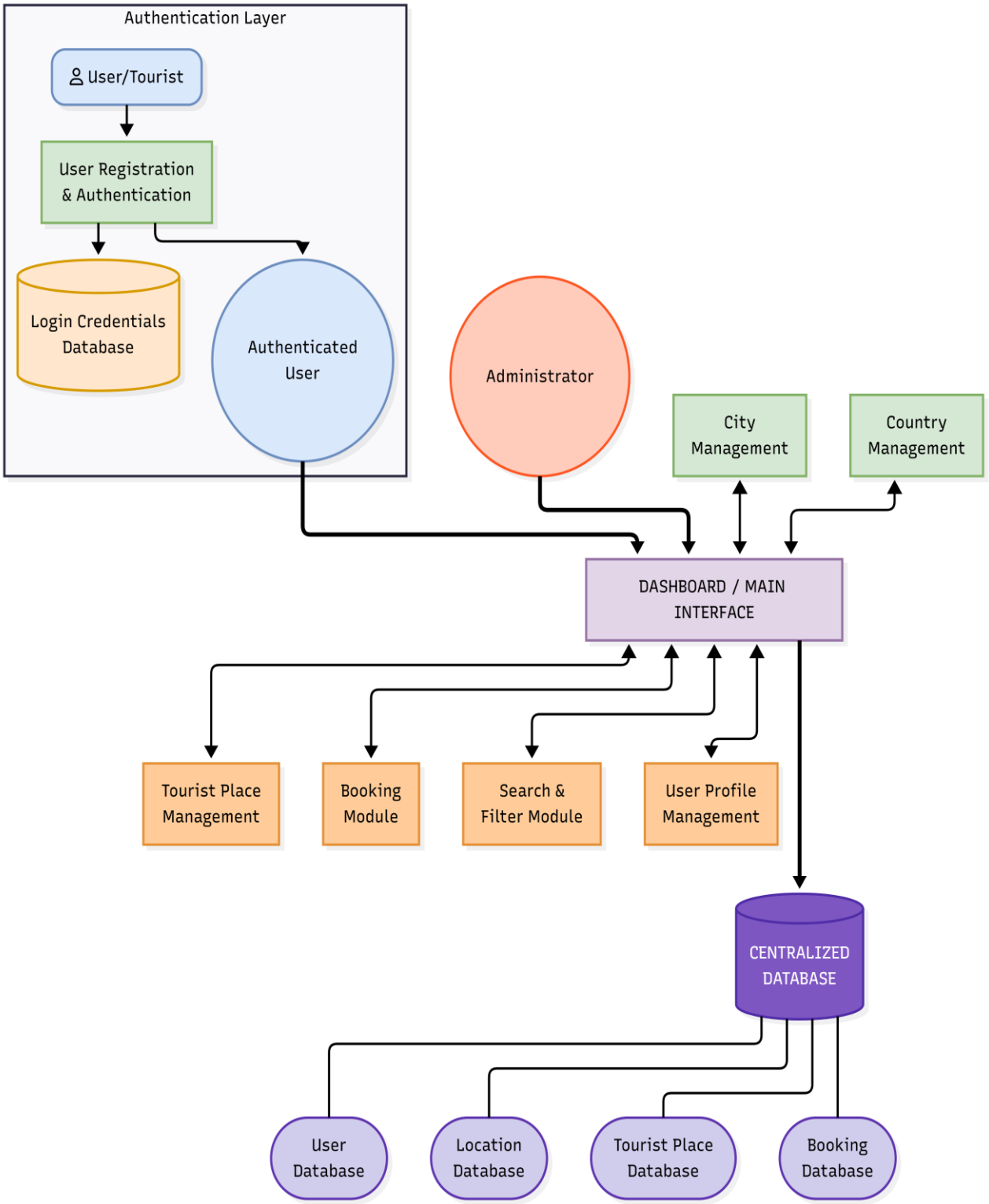
6. Search and Filter Module

Enables users to search destinations based on country, city, or category for easy navigation.

7. Database Management Module

Ensures data integrity, consistency, and efficient retrieval through structured queries and relational constraints.

System Architecture Diagram



4.FUNCTIONAL REQUIREMENTS

The ALASTKA Country-Based Tourism Information System is designed to provide organized access to tourism information while supporting efficient management of destinations, bookings, and user activities. The system fulfills the following functional requirements to ensure smooth operation, data accuracy, and an improved user experience

1. User Registration and Login

The system shall allow new users to create an account by registering with a valid email address and password. During login, users must provide correct credentials, which are verified by the system before access is granted. This authentication process ensures that only authorized users can access tourism information, make bookings, and submit reviews, thereby maintaining system security.

2. Country and City Management

Administrators shall be able to add, update, and delete country records in the system. For each country, associated cities can be created and managed. This functionality ensures that geographical tourism data remains accurate, well-organized, and up to date. The system prevents duplicate country or city entries to maintain data consistency.

3. Tourist Place Management

The system shall allow administrators to store and manage details of tourist places, including their name, category, and associated city. Tourist places are classified into categories such as **Historical**, **Natural**, and **Adventure**. This classification helps users easily understand the type of attraction and improves overall data organization within the system.

4. Destination Browsing

Users shall be able to browse available tourist destinations by selecting a country and then viewing cities within that country. From each city, users can explore the tourist places available. This step-by-step browsing approach makes it easier for users to discover attractions without requiring advanced searches.

5. Search and Filter Functionality

The system shall provide search and filter options that allow users to quickly find tourist places based on specific criteria such as country, city, or category. This feature reduces search time and improves usability by helping users directly access destinations that match their interests.

6. Booking Management

Users shall be able to create bookings for selected tourist places. Each booking is stored with a unique booking ID and the date of booking. The system maintains a clear record of all bookings, allowing users to track their reservations and administrators to manage booking data efficiently.

7. Accommodation Management

The system shall store accommodation details such as hotel or stay name, type (hotel, resort, hostel), price per night, and associated city. This information helps users plan their trips more effectively by providing accommodation options alongside tourist destinations.

8. Review and Rating System

Users shall be able to submit ratings and written reviews for tourist places they have visited. These reviews help other users make informed decisions and provide valuable feedback for improving tourism services. Each review is linked to both the user and the tourist place.

9. Administrator Dashboard

The system shall provide a centralized dashboard for administrators to view and manage user accounts, tourist places, accommodations, bookings, and reviews. This dashboard simplifies administrative tasks and enables efficient monitoring and control of system data.

10. Data Integrity and Validation

The system shall enforce data integrity by using primary key and foreign key constraints within the database. Validation rules ensure that incorrect or duplicate data entries are prevented, maintaining consistency and reliability across all records.

11. Secure Data Storage

All tourism-related information shall be stored in structured relational database tables. Secure storage practices ensure that data remains protected while allowing fast and efficient retrieval through structured queries.

12. Report Generation

The system shall support the generation of basic reports related to tourist destinations, bookings, and user activity. These reports assist administrators in analyzing system usage, monitoring tourism trends, and making informed decisions.

5.ENTITIES, RELATIONSHIPS AND ATTRIBUTES

Entity–Relationship Table

Entity 1	Relationship	Entity 2
User	A user can make multiple bookings	Booking
Booking	Each booking is created for one tourist place	TouristPlace
Country	A country contains multiple cities	City
City	A city contains multiple tourist places	TouristPlace
City	A city contains multiple accommodations	Accommodation
User	A user can write multiple reviews	Review
TouristPlace	A tourist place can receive multiple reviews	Review
TouristPlace	Tourist places are categorized as Historical, Natural, or Adventure	Category
City	Cities belong to a specific country	Country
Accommodation	Accommodations are located in cities	City

1. User

The **User** entity represents people who use the system to explore tourist destinations and make bookings.

Attributes:

- **UserID (Primary Key):** A unique ID for each user
- **Name:** Full name of the user
- **Email:** Email address used for login
- **Password:** Encrypted password for security

2. Country

The **Country** entity stores details of countries available in the tourism system.

Attributes:

- **CountryID (Primary Key):** A unique ID for each country
- **CountryName:** Name of the country

3. City

The **City** entity represents cities that belong to a specific country.

Attributes:

- **CityID (Primary Key):** A unique ID for each city
- **CityName:** Name of the city
- **CountryID (Foreign Key):** Links the city to its country

4. TouristPlace

The **TouristPlace** entity stores information about tourist attractions located in cities.

Attributes:

- **PlaceID (Primary Key):** A unique ID for each tourist place
- **PlaceName:** Name of the tourist attraction
- **Category:** Type of place (Historical, Natural, Adventure)
- **CityID (Foreign Key):** Links the place to a city

5. Booking

The **Booking** entity represents records created when users book tourist places. It depends on the User and TouristPlace entities.

Attributes:

- **BookingID (Primary Key):** Date when the booking is made
- **UserID (Foreign Key):** Links the booking to a user
- **PlaceID (Foreign Key):** Links the booking to a tourist place

6. Accommodation

The **Accommodation** entity stores details of hotels and stays available in different cities.

Attributes:

- **AccommodationID (Primary Key):** A unique ID for each accommodation
- **AccommodationName:** Name of the hotel or stay
- **Type:** Type of accommodation (Hotel, Resort, Hostel)
- **PricePerNight:** Cost per night
- **CityID (Foreign Key):** Links the accommodation to a city

7. Review

The **Review** entity stores feedback given by users for tourist places.

Attributes:

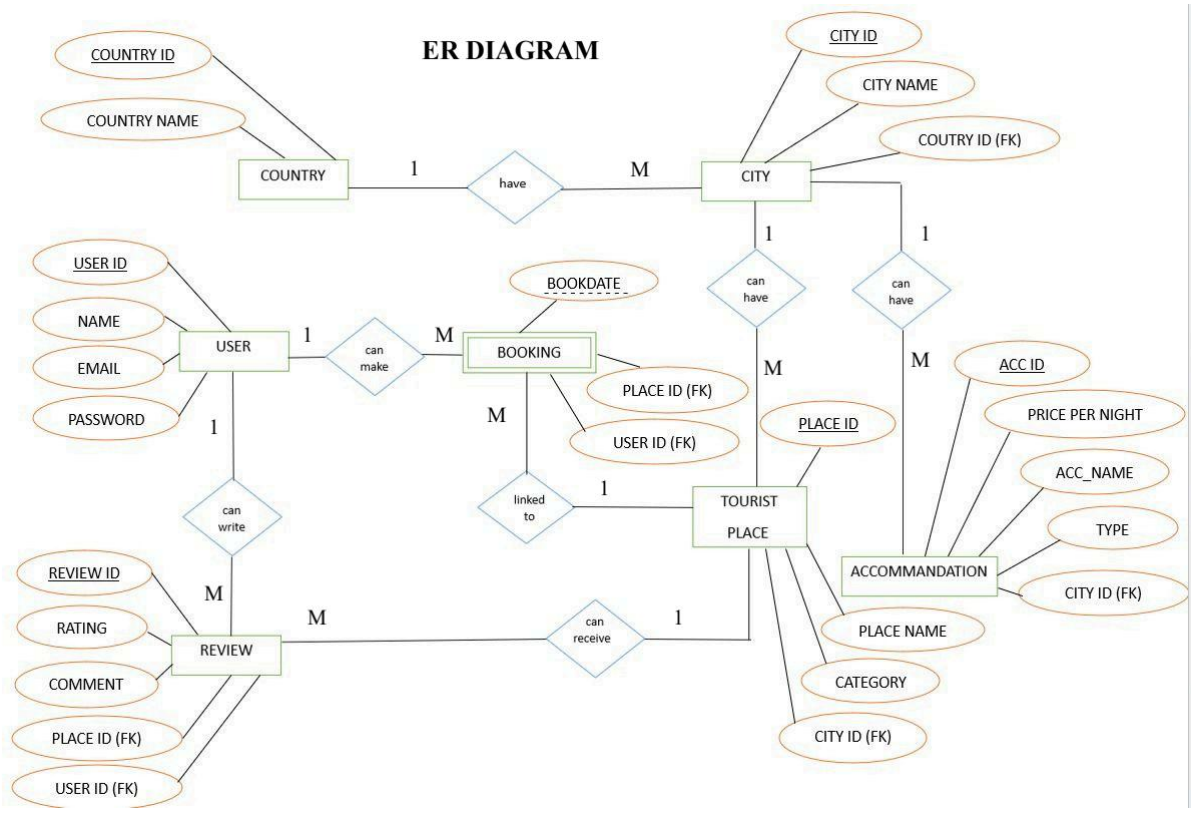
- **ReviewID (Primary Key):** A unique ID for each review
- **Rating:** Rating given by the user
- **Comment:** Feedback or review text
- **UserID (Foreign Key):** Links the review to a user
- **PlaceID (Foreign Key):** Links the review to a tourist place

5.2 **Relationships**

- One Country can have many Cities (One-to-Many).
- One City can have many Tourist Places and Accommodations (One-to-Many).
- One User can make many Bookings (One-to-Many).
- Each Booking is linked to exactly one Tourist Place (Many-to-One).
- A User can write many Reviews, and a Tourist Place can receive many Reviews. This creates a Many-to-Many relationship, which is handled using the Review entity.

5.3 **Specialization (EER Concept)**

The **TouristPlace** entity is further classified into different categories such as **Historical**, **Natural**, and **Adventure** places. This specialization helps organize tourist attractions based on their type and makes data management more efficient.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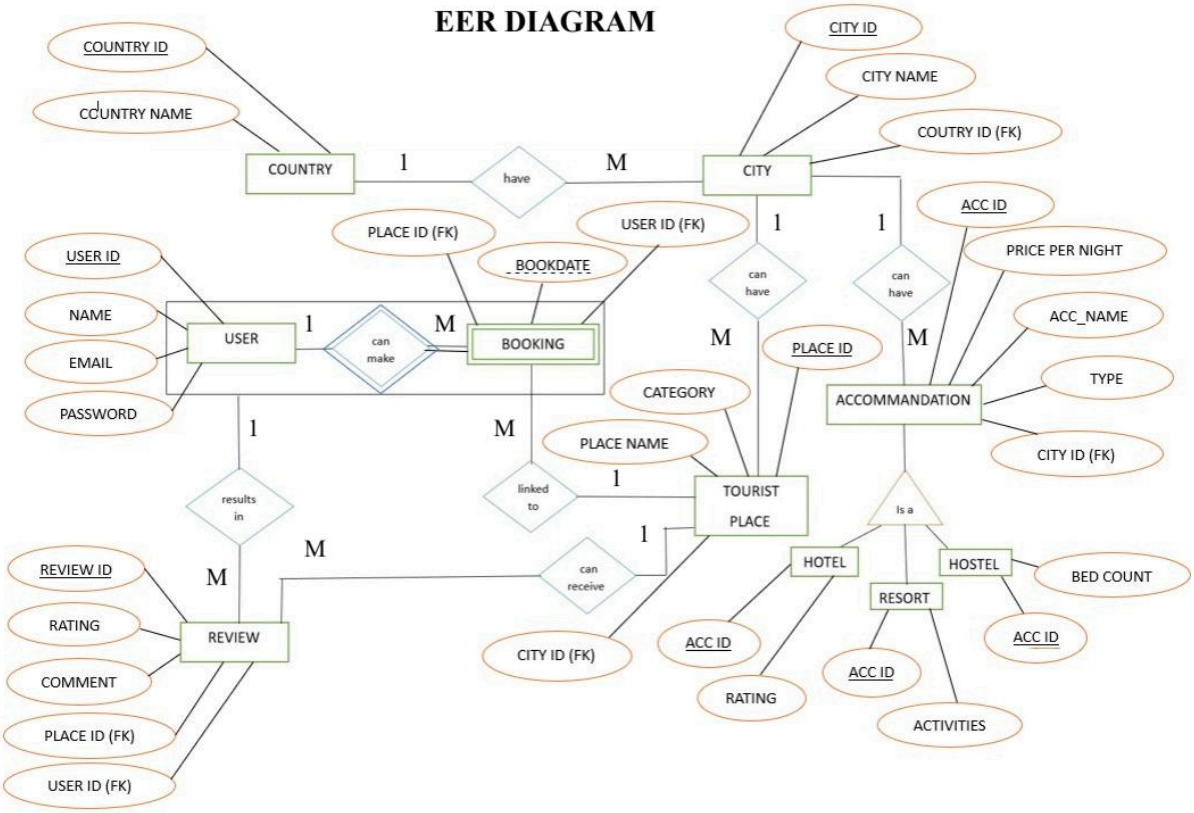
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ACTIVITIES

7. Relational Schema

*(Underlined – **Primary Key**, Italics – Foreign Key)*

User

(**UserID**, Name, Email, Password)

Country

(**CountryID**, CountryName)

City

(**CityID**, CityName, *CountryID*)

TouristPlace

(**PlaceID**, PlaceName, Category, *CityID*)

Booking

(**BookingID**, BookingDate, UserID, *PlaceID*)

Accommodation

(**AccommodationID**, AccommodationName, Type, PricePerNight, *CityID*)

Review

(**ReviewID**, Rating, Comment, *UserID*, *PlaceID*)

8. Application of Codd's Rules

The ALASTKA Country-Based Tourism Information System is designed using the **relational database model**. To ensure proper data organization, integrity, and accessibility, the system follows many of **Codd's Rules**, which define the principles of a true relational database. The application of these rules in the system is explained below.

I. Information Rule

This rule states that all information in a relational database must be stored in the form of tables, where data is represented as rows and columns.

Application in ALASTKA:
All tourism-related data in the system is stored in relational tables such as **U**ser, Country, City, TouristPlace, Booking, Accommodation, and Review.

Status: Followed
The system stores all data in a structured tabular format.

II. Guaranteed Access Rule

According to this rule, every individual data value must be accessible using a table name, primary key, and column name.

Application in ALASTKA:
Each table has a primary key, allowing users and administrators to retrieve specific data using SQL queries. For example, booking details can be accessed using the BookingID.

Status: Followed

All data elements can be uniquely and directly accessed.

III. Systematic Treatment of NULL Values

This rule ensures that the database supports NULL values to represent missing or unknown data.

Application in ALASTKA:
Optional attributes such as review comments or accommodation contact details may contain NULL values if the information is not provided.

Status: Followed
The system handles NULL values properly without affecting data integrity.

IV. Dynamic Online Catalog Rule

The database must maintain metadata (data about data), such as table definitions, constraints, and relationships, and make it accessible using the same query language.

Application in ALASTKA:
The database stores schema information such as table structures and constraints in the system catalog, which can be accessed by authorized users.

Status: Followed
Metadata is maintained and accessible through the database system.

V. Comprehensive Data Sub-Language Rule

This rule requires the use of a single, complete language to define, manipulate, and query the database.

Application in ALASTKA:
SQL is used for creating tables, inserting records, updating data, deleting records, and retrieving tourism information.

Status: Followed
SQL serves as the standard language for all database operations.

VI. View Updating Rule

This rule states that all views that are theoretically updatable should be updatable by the system.

Application in ALASTKA:
Views are mainly used for reporting and data display purposes. Complex views are not updated directly.

Status: Not Followed
View updates are limited due to system complexity and project scope.

VII. High-Level Insert, Update, and Delete Rule

The system should support operations on sets of records, not just individual rows.

Application in ALASTKA:
SQL allows bulk insertion, updating, and deletion of records, such as managing multiple bookings or reviews at once.

Status: Followed
Set-based operations are supported efficiently.

VIII. Physical Data Independence

This rule states that changes in physical storage should not affect how data is accessed by applications.

Application in ALASTKA:
Changes in storage methods or indexing do not impact application functionality or SQL queries.

Status: Followed
The system remains unaffected by physical data storage changes.

IX. Logical Data Independence

Logical changes to the database structure should not require changes to application programs.

Application in ALASTKA:
Some changes to table structures may require query or application-level modifications.

Status: Not Fully Followed
Logical schema changes can affect existing queries.

X. Integrity Independence

Integrity constraints should be defined at the database level, not in application code.

Application in ALASTKA:
Primary key and foreign key constraints are enforced directly within the database to maintain data accuracy.

Status: Followed
Data integrity rules are centrally managed by the database.

XI. Distribution Independence

The database should function the same way whether it is centralized or distributed.

Application in ALASTKA:
The system currently uses a single centralized database and does not support distributed databases.

Status: Not Followed
Distributed database implementation is outside the project scope.

XII. Non-Subversion Rule

Low-level access methods should not bypass relational integrity rules.

Application in ALASTKA:
All data access is controlled through SQL queries and database constraints, preventing unauthorized or invalid operations.

Status: Followed
Relational integrity and security are preserved.

Conclusion

The ALASTKA Country-Based Tourism Information System follows most of Codd's Rules, ensuring reliable, secure, and structured relational data management. Some rules, such as view updating and distribution independence, are not fully implemented due to the academic and limited scope of the project. Overall, the system demonstrates effective application of relational database principles.